

EXPERIMENT: 11Bitwise Operator

- i). Write a program to apply bitwise OR, AND and NOT operators on bit level.

```
#include <stdio.h>
```

```
int main () {  
    unsigned int a, b;
```

```
    printf ("Enter first number (a) : ");  
    scanf ("%u", &a);
```

```
    printf ("Enter second number (b) : ");  
    scanf ("%u", &b);
```

```
    unsigned int andResult = a & b;  
    unsigned int orResult = a | b;  
    unsigned int notA = ~a;  
    unsigned int notB = ~b;
```

```
    printf ("\n --- Bitwise Operations --- \n");  
    printf ("a & b = %u\n", andResult);  
    printf ("a | b = %u\n", orResult);  
    printf ("~a = %u\n", notA);
```

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```
printf ("%u\n", ~b);  
return 0;  
}
```

Output :- Enter first number (a): 20
Enter second number (b): 3

--- Bitwise Operations ---

$a \& b = 0$

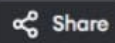
$a | b = 23$

$\sim a = 4294967275$

$\sim b = 4294967292$

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main.c



Share

Run

Output

Clear

```
1 #include <stdio.h>
2
3 int main() {
4     unsigned int a, b;
5
6     printf("Enter first number (a): ");
7     scanf("%u", &a);
8
9     printf("Enter second number (b): ");
10    scanf("%u", &b);
11
12    unsigned int andResult = a & b;
13    unsigned int orResult  = a | b;
14    unsigned int notA      = ~a;
15    unsigned int notB      = ~b;
16
17    printf("\n--- Bitwise Operations ---\n");
18    printf("a & b = %u\n", andResult);
19    printf("a | b = %u\n", orResult);
20    printf("~a   = %u\n", notA);
21    printf("~b   = %u\n", notB);
22
23    return 0;
24 }
25
```

Enter first number (a): 20
Enter second number (b): 3

--- Bitwise Operations ---

a & b = 0

a | b = 23

~a = 4294967275

~b = 4294967292

=== Code Execution Successful ===

2). Write a program to apply left shift and right shift operators.

```
#include <stdio.h>
```

```
int main () {
```

```
    int num, leftShift, rightShift;  
    int n;
```

```
    printf ("Enter a number: ");  
    scanf ("%d", &num);
```

```
    printf ("Enter number of positions to  
           shift: ");  
    scanf ("%d", &n);
```

```
    leftShift = num << n;  
    rightShift = num >> n;
```

```
    printf ("\n --- Shift operations --- \n");  
    printf ("%d << %d = %d \n", num, n, leftShift);  
    printf ("%d >> %d = %d \n", num, n, rightShift);
```

```
    return 0;
```

```
}
```

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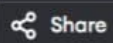
Output :- Enter a number : 19
Enter number of positions to
shift : 2

--- Shift Operations ---

$$19 < < 2 = 76$$

$$19 > > 2 = 4$$

Teacher's Signature: _____



```
1  #include <stdio.h>
2
3  int main() {
4      int num, leftShift, rightShift;
5      int n;
6
7      printf("Enter a number: ");
8      scanf("%d", &num);
9
10     printf("Enter number of positions to shift: ");
11     scanf("%d", &n);
12
13     leftShift = num << n;
14     rightShift = num >> n;
15
16     printf("\n--- Shift Operations ---\n");
17     printf("%d << %d = %d\n", num, n, leftShift);
18     printf("%d >> %d = %d\n", num, n, rightShift);
19
20     return 0;
21 }
22
```

Enter a number: 19
Enter number of positions to shift: 2

--- Shift Operations ---

19 << 2 = 76

19 >> 2 = 4

=== Code Execution Successful ===